Computational Text Analysis
IDHEAP Doctoral School

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Unil.

The plan for today

1. Using Simple Methods well: Martin (2018)
2. Recap of Day 1 & dictionaries
3. Script 4 continued: Dictionaries
4. Input: Machine learning & topic models
5. Script 5: Topic models
6. In-class project and pitch

What is one thing you learned yesterday? What is still unclear?

Martin (2018): How Simple Methods Can Trump Complex Ones

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Word Frequencies: Martin 2018

IMAGINE ALL THE PEOPLE
Literature, Society, and Cross-National Variation in Education Systems

By CATHIE JO MARTIN*

Who read the text?

Group Work: Martin (2018)

- Reopen the text and answer the questions below.
- First think about them on your own and get back into the text (10 minutes). Then you will work together in a group to discuss the text (10 minutes).

1. What is the research question of Martin (2018)?
2. What text data does she use?
3. Who created the text data? What does this mean for her analysis?
4. How does she analyse the text?
5. What does she find?
6. Do you think this is a good research design? Is the analysis convincing?

Martin (2018): Question and data

1. Research question:

- Why did poor, agrarian Denmark create mass public schooling early (1814) and a vocational track, while rich, industrial Britain waited until 1870 and chose uniform academic secondary schools?
- Her answer to test: literary narratives about education, the individual and society shape education policy choices

2. Text data:

- 562 British and 521 Danish works of fiction (novels, poems, plays), 1700–1920

3. Who created the data?

- Canonical authors, writing for (and read by) educated elites, later mass readers

Martin (2018): Methods and findings

4. **Methods:**
   - **Dictionaries**: hand-made word lists (e.g. individualism, society), counted only when around education words
   - **LDA topic modelling (unsupervised)**
   - **Close reading**: 4 matched pairs of British and Danish novels (e.g. Defoe, Robinson Crusoe vs Holberg, Niels Klim's Journey Under the Ground)

5. Findings: Education words rise before education reforms (Denmark 1814, Britain 1870)

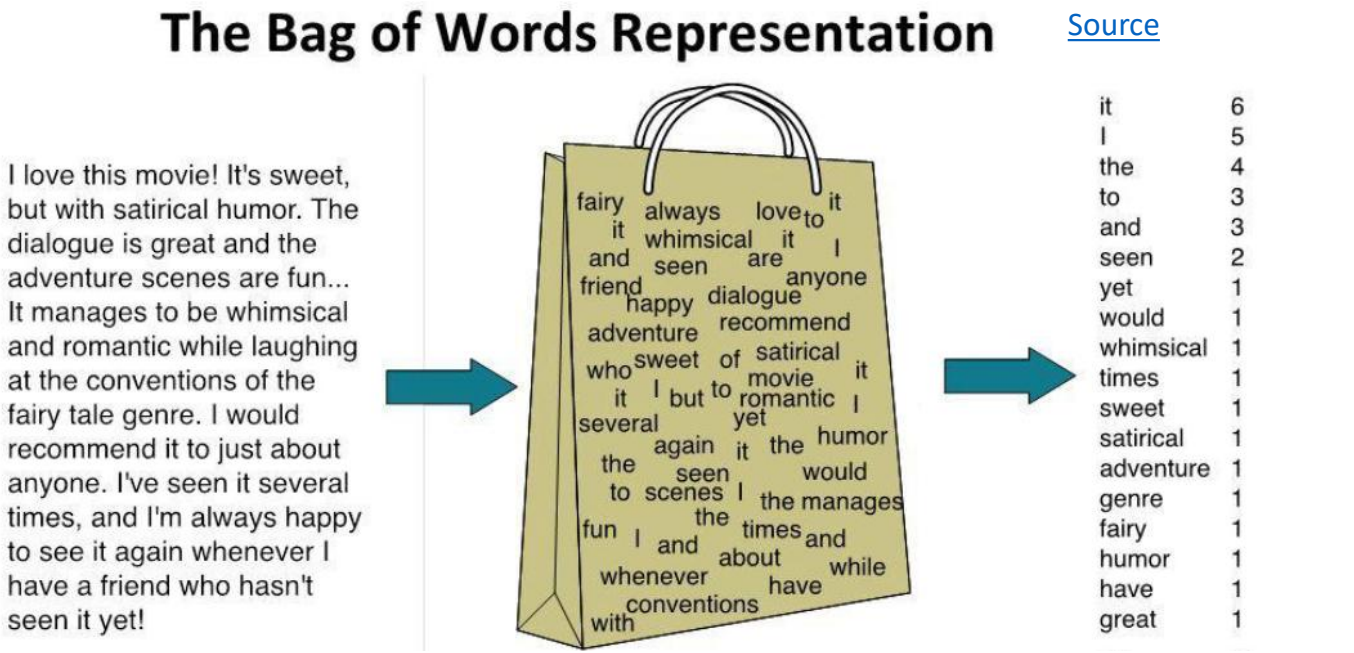
- British education passages: more individualism and upper-class words; Danish: more society and (1770–1870) governance words; convergence after 1870
- LDA 1720–1770: British topics are on knowledge, mind, gentlemen, ladies; Danish on truth, Latin, king, God, church

Lessons from Martin (2018):

1. Find the right method, not the most hyped / novel!
   - The right method depends on your research question and available data
2. No need to choose sides on the qualitative/quantitative divide. We can gain great leverage from using both.
3. Be open about possible data sources.

Recap of Day 1

Recap: Basic Assumption and the Bag of Words Representation



Document → Tokens → Document-Feature-Matrix

The workflow

- Install & load the packages you will use
- Acquire documents and import them into RStudio
- Pre-Processing
  - Tokenize
  - Create document-feature-matrix
  - Learn about data at hand
  - Remove non-text (punctuation, symbols, numbers)
  - Remove meaningless words (stopwords, potentially trim)
  - Possibly stem
- Choose method for text analysis
- Plot and visualise

Advanced Transformations: Create a subset / group features

- Sometimes we want to analyse only some part of the data
- Therefore, we need to create a subset of the data

```
nogermany_system_dfm <- dfm_subset(systems_dfm,country!="Germany")
Germany_dfm <- dfm_subset(systems_dfm, country == "Germany")
```

Topfeatures

- Before pre-processing:

the	of	and	in	to	education	for	a	is	are
408922	251546	218485	138468	117839	105486	99107	86702	58311	47690

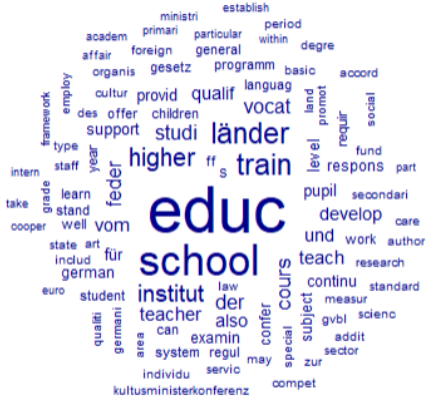
- After pre-processing:

educ	school	student	institut	train	programm
125054	71368	27918	27828	25322	23814
teacher	higher	year	provid		
23326	23166	22670	19144		

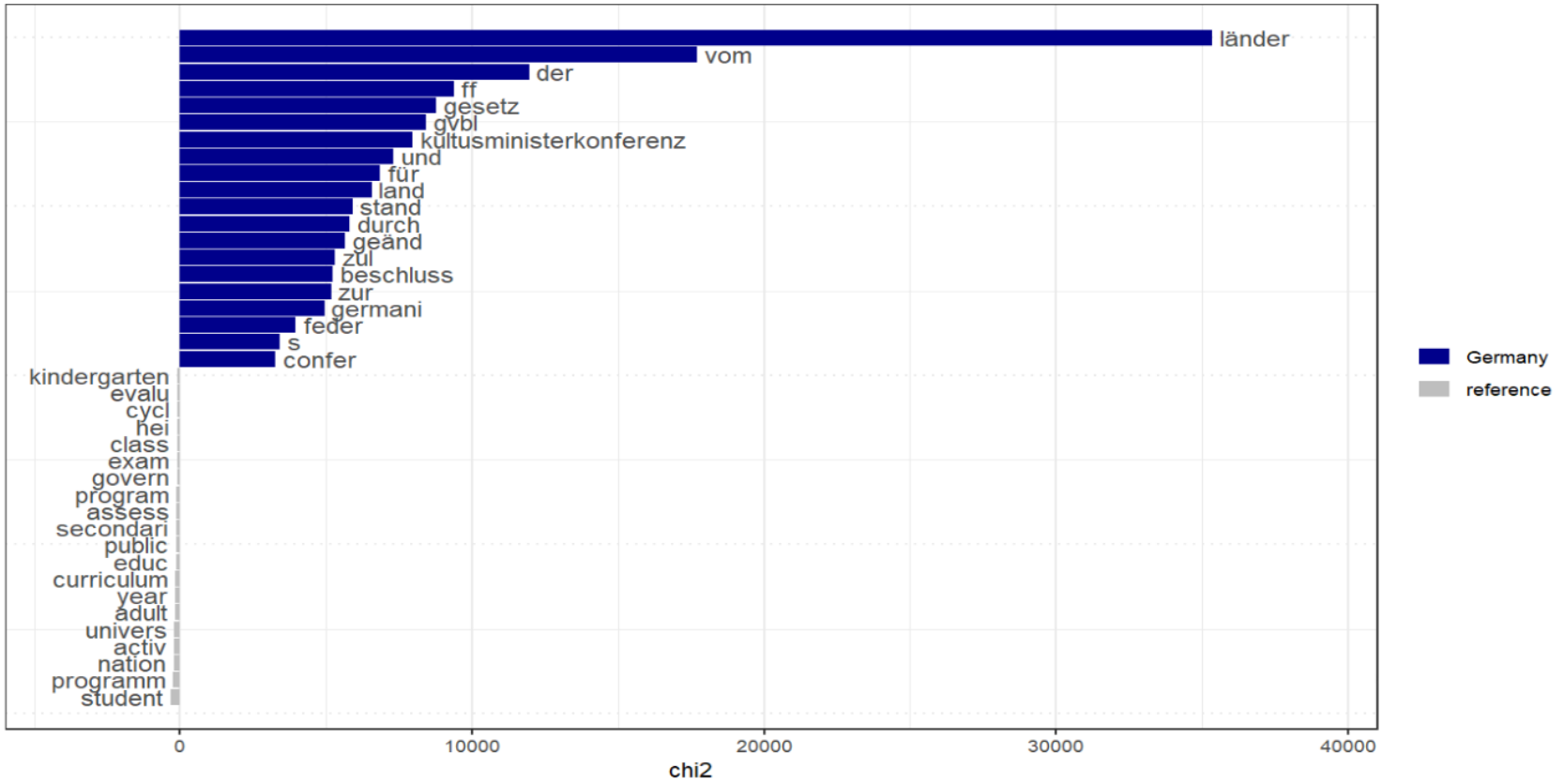
Wordclouds

```
textplot_wordcloud(systems_dfm, max_words = 100)
```





Plotting and interpreting keyness



Dictionaries

- **Keys** that stand for a theorized concept
- **Values** that represent the keys empirically

- Example: Food (key); bread, apple, banana, chocolate, potatoes (values)

Workflow: Working with existing dictionaries – Step 1: Load data and Pre-Process

1. 2. 3. 4. 5. & 6.

```
debate_data <- read.csv("C:/Users/.../us_election_2020_1st_presidential_debate.csv")


corp_debate <- corpus(debate_data)

toks_debate <- tokens(corp_debate,  remove_punct=T,  remove_numbers=T, remove_symbols=T)
dfm_debate <- dfm(toks_debate)
dfm_debate <- dfm_remove(dfm_debate,stopwords("en"))
dfm_debate <- dfm_wordstem(dfm_debate,"en")
```

What do we do when running the code lines 1-6?

Workflow: Dictionaries – Step 2: Load Dictionary

- 1: Save dictionary in working directory
- 2: Load with dictionary function
- Here: Newsmap geography dictionary

```
newsmap_dict <- dictionary(file = "english.yml",
                           format = "YAML")
```

```
newsmap_dict
```

```
## Dictionary object with 5 primary key entries and 3 nested levels.
## - [AFRICA]:
##   - [EAST]:
##     - [BI]:
##       - burundi, burundian*, bujumbura
##     - [DJ]:
##       - djibouti, djiboutian*
##     - [ER]:
##       - eritrea, eritrean*, asmara
##     - [ET]:
##       - ethiopia, ethiopian*, addis ababa
##     - [KE]:
##       - kenya, kenyan*, nairobi
##     - [KM]:
##       - comoros, comorian*, moroni
##     [ reached max_nkey ... 13 more keys ]
##   - [MIDDLE]:
##     - [AO]:
##       - angola, angolan*, luanda
##     - [CD]:
##       - democratic republic congo, dr congo, drc, democratic republic congolese, dr congolese, kinshasa
```

Workflow: Dictionaries – Step 3: Apply Dictionary

- Use dfm_lookup() command to match dictionary words with dfm

```
dict_dfm_results <- dfm_lookup(dfm_debate,newsmap_dict)
dict_dfm_results[610:615,111:119]
```

```
## Document-feature matrix of: 6 documents, 9 features (96.30% sparse) and 2 docvars.
##          features
## docs      AMERICA.NORTH.GL AMERICA.NORTH.PM AMERICA.NORTH.US ASIA.CENTER.KG
##   text610                0                0                2               0
##   text611                0                0                0               0
##   text612                0                0                1               0
##   text613                0                0                0               0
##   text614                0                0                0               0
##   text615                0                0                0               0
##          features
## docs      ASIA.CENTER.KZ ASIA.CENTER.TJ ASIA.CENTER.TM ASIA.CENTER.UZ
##   text610              0              0              0              0
##   text611              0              0              0              0
##   text612              0              0              0              0
##   text613              0              0              0              0
##   text614              0              0              0              0
##   text615              0              0              0              0
##          features
```

Workflow: Dictionaries – Step 4: Frequency analysis

- Use textstat_frequency() command to get the frequency and a ranking of the prevalence of regions

```
dict_dfm_results %>% textstat_frequency()
```

```
##              feature frequency rank docfreq group
## 1   AMERICA.NORTH.US        44    1      35   all
## 2       ASIA.EAST.CN        10    2       9   all
## 3     EUROPE.EAST.RU         6    3       6   all
## 4      ASIA.SOUTH.IN         2    4       2   all
## 5  AMERICA.CENTER.MX         1    5       1   all
## 6   AMERICA.SOUTH.BR         1    5       1   all
## 7       ASIA.EAST.JP         1    5       1   all
## 8       ASIA.WEST.IQ         1    5       1   all
## 9    EUROPE.NORTH.IE         1    5       1   all
## 10    EUROPE.WEST.DE         1    5       1   all
```

Dictionary creation: Best practice

1. Creating a top-features list
   - From data
2. Deductively selecting features
   - Which of these explain the theoretical concept well (unambiguously)
3. Manual validation
   - Human coders determine whether feature should be included or not



Political Communication

ISSN: 1058-4609 (Print) 1091-7675 (Online) Journal homepage: https://www.tandfonline.com/loi/upcp20

(Re)Claiming Our Expertise: Parsing Large Text Corpora With Manually Validated and Organic Dictionaries

Ashley Muddiman, Shannon C. McGregor & Natalie Jomini Stroud

Now you: Script 4 – continued

- Continue with “4_Dictionaries” where you stopped yesterday
- Create your own dictionary (“dictionary creation”)
- Fast? Try the optional part with the Eurydice data

Supervised and Unsupervised Machine Learning for Text Analysis: Focus on Topic Modelling

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Computational Text Analysis and Human Involvement

- (Computational) Text Analysis requires different levels of human involvement depending on the selected method
- 100% human involvement: Qualitative Content Analysis
- 1-99% human involvement: Supervised methods (dictionaries, supervised machine learning, semi-supervised machine learning)
- (Almost) no human involvement (unsupervised machine learning)

Machine Learning

- “A computer program is said to learn from **experience E** with respect to some class of **tasks T** and **performance measure P** if its performance at tasks in T, as measured by P, improves with **experience E**” (Tom Mitchell, Machine Learning, 1997)
- In text analysis generally used for classification tasks
- Task T = classification of texts
- Experience E = pre-classified text
- Performance measure P = correctly classified texts

Machine Learning: Classifiers

- We learn which text features predict categories of interest through classifiers (algorithms)
- Logic is simple and similar to regressions:
  - Regressions: Does the number of universities predict a highly skilled population?
  - Dictionary: Does the word ‘university’ predict a highly skilled population?
  - Classifier: Does the word ‘university’ predict a highly skilled population? Does the word ‘house’ predict a highly skilled population? Does the word…?

Machine learning in text-as-data

- **Generalization**: Classifiers learn how to correctly predict output from known inputs but can also classify unknown data.
- **Overfitting**: By learning from data, it might work very well for the known sample, but not so well for unseen data.
- Always have in mind what data a model was trained with when interpreting results.

Machine learning in text-as-data

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Machine learning in text-as-data

- **Unsupervised Machine Learning:** Requires no prior information, training set, labelled texts or seed words
- **Supervised Machine Learning:** Some degree of human involvement. Often, we label data to train a model and then use it to classify large amounts of data
- Today: **Topic models (**supervised and unsupervised)

Topic Models

- Topic models are algorithms for discovering the main "themes" in an unstructured corpus
- What is a topic?
  - Google: “a matter dealt with in a text, discourse, or conversation; a subject”
  - Topic model definition: a probability distribution over a fixed word vocabulary
- Practically this means topics are made up of words. Each word has a probability that estimates how likely it is to belong to a certain topic.

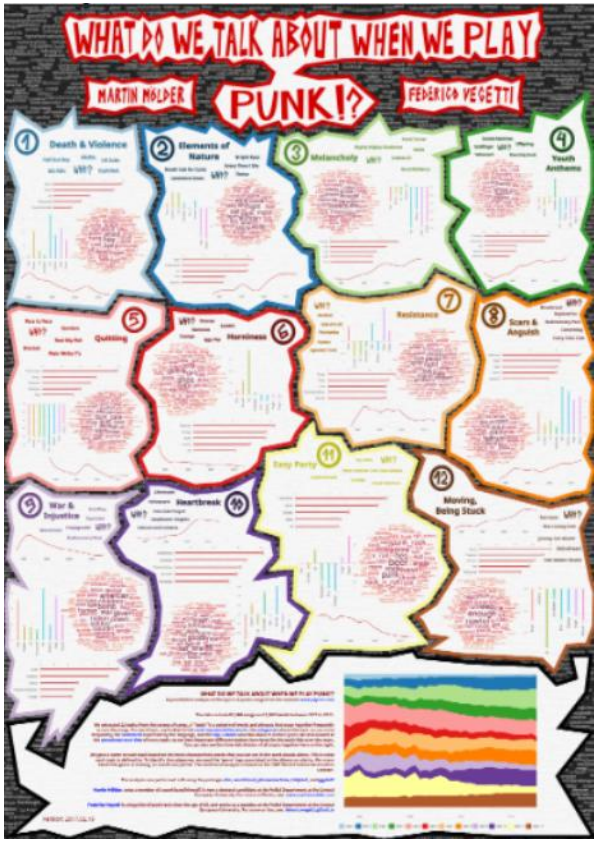
Topic Models: Example

- Imagine a corpus that includes the following words
  - Gene, dna, genetic, data, number, computer
- Texts about genetics will more frequently use the words gene, dna and genetic, but less frequently use data, number and computer and the other way around.
- Probability of each word belonging to either the topic genetics or computation:

Topic	gene	dna	genetic	data	number	computer
Genetics	0.4	0.25	0.3	0.02	0.02	0.01
Computation	0.02	0.01	0.02	0.3	0.4	0.25

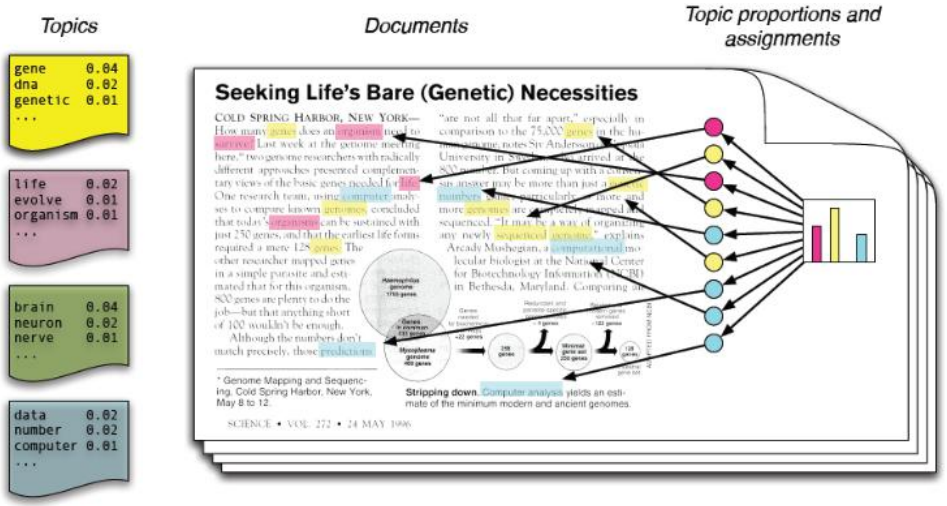
Topic Models: Typical use

- To study trajectories of large topics
  - In newspapers, scientific journals, archives, social media etc.
- To study different frames of the same debate
  - Highlighting different topics, using different words
- To get a first impression of large amounts of data
- One example: Punk lyrics!



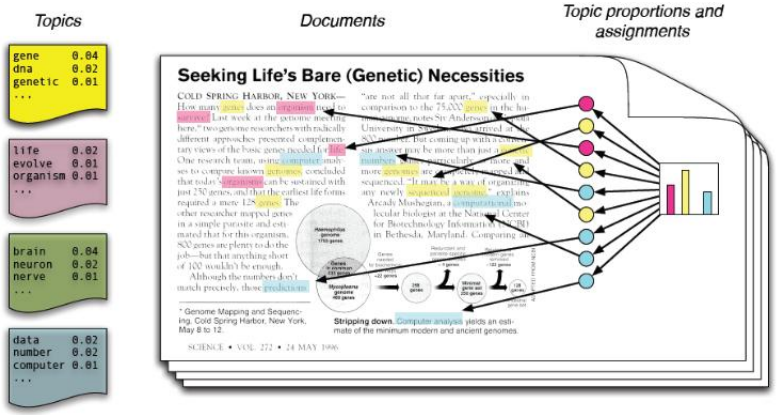
Topic Models: Latent Dirichlet Allocation (LDA)

- Each **topic** is a distribution over words
- Each **document** is a mixture of corpus-wide topics
- Each **word** is drawn from one of those topics



Topic Models: Latent Dirichlet Allocation (LDA)

- LDA topic modelling has two goals:
  1. For each document, allocate its words to as few topics as possible.
  2. For each topic, assign high probability to as few words as possible.
- The model optimizes this trade-off
- Can be unsupervised or supervised (seeded). We will learn about both.



LDA Topic Models: Workflow

- First step: Selecting data set, loading data and pre-processing
- We work with the Eurydice data again in the presentation as you already know the data
- What are the steps of pre-processing?

```
eurydice <- readRDS("C:/User[redacted]

corp_eurydice <- corpus(eurydice)

toks_eurydice <- tokens(corp_eurydice,  remove_punct=T,  remove_numbers=T, remove_symbols=T)

dfm_eurydice <- dfm(toks_eurydice)
dfm_eurydice <- dfm_remove(dfm_eurydice,stopwords("en"))
dfm_eurydice <- dfm_wordstem(dfm_eurydice,"en")
```

LDA Topic Models: Workflow

- Creating topic models without further supervision requires just two decisions:

1. How do I need to clean the data?
2. How many topics should my model predict?
    - There are different ways to determine the optimal number of topics. We will table this question for now to keep it simple.

LDA Topic Models: Workflow

- Technical process: Run following command
  - x <- textmodel_lda(x, k = x)
  - What do the different x stand for?

LDA Topic Models: Workflow

- Technical process: Run following command
  - x <- textmodel_lda(x, k = x)
  - What do the different x stand for?

```
# run the model. This will take time
tmod_lda <- textmodel_lda(dfm_eurydice, k = 10)
```

LDA Topic Models: Workflow

- The model determines the optimal distribution of words over topics for the given dfm.
- The first output we get is a list of topics and the words that with a high probability predict that a text belongs to a topic.
- We can get a list of terms associated with each topic to see if it makes sense.

```
# what are the terms associated with each topic
terms(tmod_lda, 20)
```

Topic Models

- What can we learn about the data from this unsupervised topic modelling?

	topic1	topic2	topic3	topic4	topic5	topic6
[1,]	"teacher"	"school"	"educ"	"educ"	"educ"	"educ"
[2,]	"staff"	"educ"	"institut"	"train"	"de"	"develop"
[3,]	"school"	"pupil"	"school"	"vocat"	"institut"	"european"
[4,]	"teach"	"secondari"	"qualiti"	"adult"	"access"	"project"
[5,]	"work"	"year"	"evalu"	"qualif"	"school"	"mobil"
[6,]	"profession"	"student"	"higher"	"learn"	"republ"	"languag"
[7,]	"educ"	"subject"	"law"	"provid"	"o"	"programm"
[8,]	"employ"	"class"	"govern"	"cours"	"czech"	"countri"
[9,]	"head"	"special"	"ministri"	"programm"	"slovak"	"intern"
[10,]	"servic"	"primari"	"respons"	"develop"	"avail"	"nation"
[11,]	"year"	"grade"	"nation"	"profession"	"act"	"cooper"
[12,]	"manag"	"teach"	"council"	"level"	"la"	"support"
[13,]	"appoint"	"teacher"	"public"	"skill"	"et"	"student"
[14,]	"may"	"languag"	"system"	"employ"	"du"	"activ"
[15,]	"salari"	"curriculum"	"bodi"	"compet"	"isc"	"new"
[16,]	"train"	"assess"	"establish"	"institut"	"des"	"increas"
[17,]	"requir"	"general"	"regul"	"system"	"v"	"learn"
[18,]	"posit"	"compulsori"	"state"	"guidanc"	"provid"	"particip"
[19,]	"condit"	"upper"	"assur"	"centr"	"facil"	"cultur"
[20,]	"period"	"need"	"develop"	"work"	"last"	"strategi"

	topic7	topic8	topic9	topic10
[1,]	"educ"	"higher"	"feder"	"educ"
[2,]	"school"	"student"	"canton"	"school"
[3,]	"provid"	"studi"	"der"	"children"
[4,]	"learn"	"educ"	"länder"	"fund"
[5,]	"support"	"programm"	"und"	"year"
[6,]	"develop"	"univers"	"für"	"provid"
[7,]	"includ"	"institut"	"tel"	"public"
[8,]	"govern"	"degre"	"austria"	"institut"
[9,]	"qualif"	"cours"	"vom"	"age"
[10,]	"skill"	"academ"	"austrian"	"privat"
[11,]	"level"	"research"	"german"	"support"
[12,]	"framework"	"year"	"univers"	"parent"
[13,]	"fund"	"qualif"	"liechtenstein"	"grant"
[14,]	"need"	"requir"	"websit"	"municip"
[15,]	"act"	"examin"	"confer"	"child"
[16,]	"colleg"	"scienc"	"die"	"care"
[17,]	"nation"	"level"	"s"	"fee"
[18,]	"set"	"teach"	"swiss"	"kindergarten"
[19,]	"provis"	"doctor"	"educ"	"state"
[20,]	"also"	"profession"	"last"	"famili"

Topic Models

- Topic 1 seems to be about the **teaching profession**
- Topic 2 seems to be about **school education** but there is room for ambiguity
- Topic 3 is about **governance**
- Topic 4 is about education as a means to create **human capital on the labor market**
- Topic 6 seems to be **European integration**
- Topic 8 is about **higher education**.
- Topic 9 is about **federalism**
- Topic 10 seems to be about the perspective of **families** and early childhood education and care (**ECEC**)
- Topics 5 and 7 are hard to interpret because of language diversity in the data set. Lots of gibberish and very frequent words. Common denominator: It is about **education**. More cleaning could help.

	topic1	topic2	topic3	topic4	topic5	topic6
[1,]	"teacher"	"school"	"educ"	"educ"	"educ"	"educ"
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[8,]	"govern"	"degre"	"austria"	"institut"
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LDA Topic Models: Workflow

- Let's get a list of the prevalent topics of the first 20 documents using the command 'head(topics(x), 20)'
- Many of the first texts are classified as topic 7 which was a mixture of general education words (education, school, learn*, provid*, support*, develop*).

```
# get prevalent topics of first 20 documents (texts)
head(topics(tmod_lda), 20)
```

```
##  text1  text2  text3  text4  text5  text6  text7  text8  text9 text10 text11
## topic7 topic7 topic6 topic7 topic7 topic6 topic6 topic1 topic6 topic6 topic6
## text12 text13 text14 text15 text16 text17 text18 text19 text20
## topic6 topic6 topic2 topic2 topic6 topic9 topic2 topic2 topic2
## 10 Levels: topic1 topic2 topic3 topic4 topic5 topic6 topic7 topic8 ... topic10
```

LDA Topic Models: Workflow

- We can also count how prevalent the topics are in the overall corpus by counting how many times a topic is predicted to be dominant in a document
- By far the most relevant topics are school education (topic 2) and skills / education as human capital (topic 4), followed by European integration (topic 6).

topic1	topic2	topic3	topic4	topic5	topic6	topic7	topic8	topic9	topic10
402	833	494	728	89	612	349	484	36	429

LDA Topic Models: Workflow

- To get these results we need to integrate the dominant topic of each document into the document-feature matrix or database
- We assign a new variable ‘topic’ to the existing dfm and define that the content of this new variable is the topics of our lda model.
- With the table function we get the frequency of each topic
- Use the $ between dfm and variable name to specify a variable of the dfm

```
# assign topic as a new document-level variable
dfm_eurydice$topic <- topics(tmod_lda)

# cross-table of the topic frequency
table(dfm_eurydice$topic)
```

LDA Topic Models in Martin (2018): Are British writers more individualist and Denmark more collective?

TABLE 2
TOPICS IN BRITAIN AND DENMARK 1720–1770[a]

Britain

1. learn, knowledge, word, nature, man, life, great, mind, part, men, world, education, prince
2. read, history, great, book, lady, chap, Dr., trim, knowledge, Mrs., work
3. word, book, hand, write, uncle, read, count, half, head, eye, turn, world
4. word, history, great, man, good, never, lady, think, thought, miss, give, letter, write, answer, book
5. write, read, dear, sir, good, lady, letter, hut, word, think, give, hope, love, happiness

Denmark

1. people, learn, meaning, truth, give, wise, right, word, wild/lost, hand, hold, rector
2. discuss, part, Greek, writing, translate, Latin, book, philosophy, learning, indicate
3. man, wild/lost, right, same/together, old, number, people, hold, calling, learning, learn
4. king, fault, learn, God, church, hold, right, less, follow, give, love, wise
5. hand, poem, learning, name, Latin, man, philosophy, book, write, rector

Seeded LDA Topic Models

- In seeded LDA topic modelling, you can predefine topics using a dictionary of "seed words".
- But we only nudge the model: Jagarlamudi et al. (2012: 205):

"importantly, we only encourage the model to follow the seed sets and do not force it. So if it has compelling evidence in the data to overcome the seed information then it still has the freedom to do so."

- More on the theory of seeded LDA: Watanabe & Zhou (2022): Theory-Driven Analysis of Large Corpora: Semisupervised Topic Classification of the UN Speeches

Seeded LDA Topic Models: Workflow

- First, we need a dictionary. You can create it just like in the dictionary analysis.
- Example in Eurydice: Public or private governance in education. Are state actors or private companies more important in education?

Seeded LDA Topic Models: Workflow

- This dictionary is an incomplete mixture of terms from the Lexicoder Policy Agendas Dictionary and frequent words from the text that are (relatively) unambiguous.

```
# Define a dictionary
dict_PubvPriv <- dictionary(list(public = c("public","govern*", "state","regulat*", "control",
"nation*", "assist", "benefit", "care"), private = c("privat*","enterpr*", "firm*","industr*",
"work*", "marke*","competit*", "dereg*", "autonom")))

print(dict_PubvPriv)
```

```
## Dictionary object with 2 key entries.
## - [public]:
##   - public, govern*, state, regulat*, control, nation*, assist, benefit, care
## - [private]:
##   - privat*, enterpr*, firm*, industr*, work*, marke*, competit*, dereg*, autonom
```

Seeded LDA Topic Models: Workflow

- Run the seeded LDA topic model (slda) using the textmodel_seededlda() command.
  - Define the name of the model
  - Define the dfm the model uses
  - Define the dictionary that provides the seed words. The number of topics is defined by the number of topics in your dictionary. In the example it is only two. Usually there are more topics in texts.

```
tmod_slda <- textmodel_seededlda(dfm_eurydice, dictionary = dict_PubvPriv)
terms(tmod_slda, 20)
```

Seeded LDA Topic Models: Workflow

- With the command "terms(tmod_slda, 20)" we can again see the 20 top terms for each topic.

```
      public     private
 [1,] "educ"     "school"
 [2,] "nation"   "educ"
 [3,] "public"   "work"
 [4,] "higher"   "teacher"
 [5,] "govern"   "year"
 [6,] "institut" "student"
 [7,] "state"    "pupil"
 [8,] "train"    "secondari"
 [9,] "programm" "teach"
[10,] "univers"  "children"
[11,] "develop"  "studi"
[12,] "student"  "privat"
[13,] "adult"    "train"
[14,] "provid"   "programm"
[15,] "fund"     "subject"
[16,] "qualiti"  "provid"
[17,] "learn"    "special"
[18,] "qualif"   "level"
[19,] "research" "vocat"
[20,] "support"  "primari"
```

Seeded LDA Topic Models: Workflow

- Two topics can be too few. Example: The topic private is captured by the much more prevalent topic of school education.
- Second concept (public) seems to be captured quite well.
- This is a warning: The model does not really capture public versus private topics. Therefore, the method is not useful for the question at hand.

```
      public     private
 [1,] "educ"     "school"
 [2,] "nation"   "educ"
 [3,] "public"   "work"
 [4,] "higher"   "teacher"
 [5,] "govern"   "year"
 [6,] "institut" "student"
 [7,] "state"    "pupil"
 [8,] "train"    "secondari"
 [9,] "programm" "teach"
[10,] "univers"  "children"
[11,] "develop"  "studi"
[12,] "student"  "privat"
[13,] "adult"    "train"
[14,] "provid"   "programm"
[15,] "fund"     "subject"
[16,] "qualiti"  "provid"
[17,] "learn"    "special"
[18,] "qualif"   "level"
[19,] "research" "vocat"
[20,] "support"  "primari"
```

Seeded LDA Topic Models: Workflow

- You can see which topic is prevalent in which document with the topics() command.
- With the head(topics(x), 20) command you get a glimpse at the topics of the first 20 documents

```
topics_slda <- topics(tmod_slda)

head(topics(tmod_slda), 20)
```

```
##   text1   text2   text3   text4   text5   text6   text7   text8   text9  text10
##  public  public  public  public  public  public  public  public  public  public
##  text11  text12  text13  text14  text15  text16  text17  text18  text19  text20
##  public  public  public private  public  public  public  public private private
## Levels: public private
```

Seeded LDA Topic Models: Workflow

- To compare the prevalence of the topics on the dfm, you can add the topics as variable to your dfm and then apply the table(dfm$topic) command.

```
dfm_eurydice$topic <- topics(tmod_slda)

table(dfm_eurydice$topic)
```

```
##
##  public private
##    2261    2195
```

Seeded LDA Topic Models: Best practice

- What can seeded topic modelling deliver?
- Example from Watanabe & Zhou (2022), “Theory-Driven Analysis of Large Corpora: Semisupervised Topic Classification of the UN Speeches.”
- What do we see in this figure?

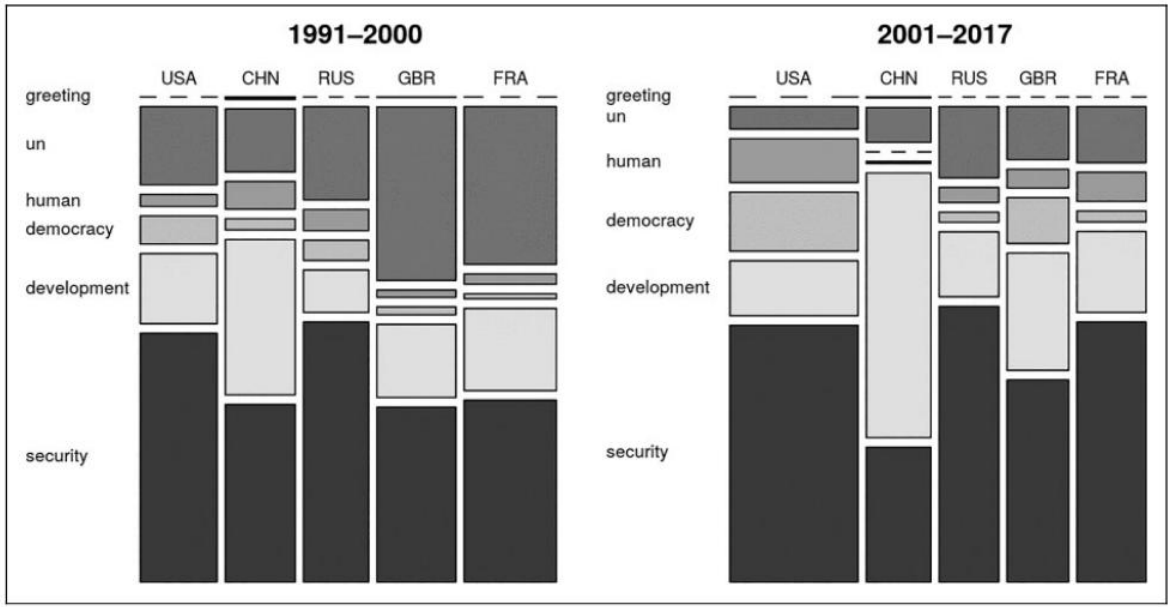


Figure 8. Topic in speeches by the Security Council permanent members before and after the 9/11 attack. *Note*. Width indicates length of speeches in number of sentences.

Seeded LDA Topic Models: Best practice

- Changes in General Assembly speeches of the Security Council's permanent members after 9/11:
  - US speeches become longer compared to other members
  - Security topic becomes more central among the speeches of the USA, GBR and FRA, but clearly less salient in the Chinese speeches

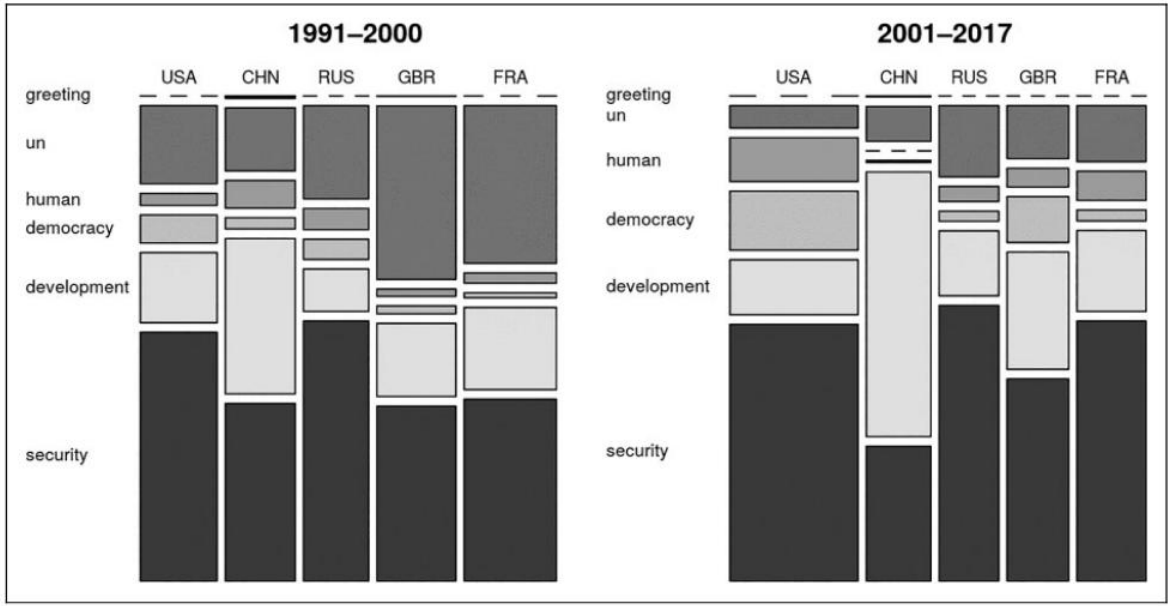


Figure 8. Topic in speeches by the Security Council permanent members before and after the 9/11 attack. *Note*. Width indicates length of speeches in number of sentences.

Discuss with your neighbour

Discussion in pairs, then we share in plenary

1. What is the difference between supervised and unsupervised machine learning? How does this affect topic modelling?
2. What are the decisions we have to take when we do seeded LDA topic modelling?
3. What are the advantages / disadvantages of the different methods we have learned?

Now you! Work on Script 5 Topic Models

- LDA first, then seeded LDA
- Script “5_Topic Models” with the Eurydice data

In-Class Project

Recent applications for inspiration

Kittel (2025) · *Government and Opposition*
Simply Speaking? Language Complexity among (Non-)Populist Actors in Parliamentary Debates

Hurka, Knill & Steinebach (2024) · *Regulation & Governance*
Rules as policy data? Measuring and linking policy substance and legislative context

Thies (2025) · *Regulation & Governance*
State Intervention in Vocational Education: Training for the Digital and Green Transitions

Carstensen, Ibsen, Jensen & Lund-Sørensen (2025) · *Regulation & Governance*
Mapping Green Skills in Collective Skill Formation Systems: A Natural Language Processing Analysis of Danish Vocational Education and Training

In-Class Project

- This could be part of your PhD or a paper
- … or it could be just to practice what you have learned.
- The general idea: To really learn CTA, I believe you must apply it, even if only in simple forms at first.
    - This is how I learned it.
- You can bring your own data or use the data I provide
- Do not worry if you have not understood everything yet. You have several hours and my assistance.

Your task:

- This is not just about coding. This is about applying new methods in a research context. Therefore:

1. Define a research question that has a theoretical foundation (remember the first day or remember why you are interested in your topic).
2. Import, clean and organize your data. This might be the biggest challenge for those with their own data.
3. Apply those methods you think are helpful to answer your research question.
    - This can be simple, like wordclouds and topfeatures or more advanced, including topic models
4. Interpret the findings: What have you learned?
5. Visualize your findings in a graph or a nicely knitted R Markdown file if you can. Careful, knitting might take too long if you use topic modelling. Prepare a short presentation of 3 minutes.

Text data to work with

- **Your own data**
- **UN General Debates:** countries' annual UN General Assembly speeches, 1970–2015, 7,507 speeches. Variables: session, year, country, text.
- **Brexit Bill:** House of Commons debates on the EU Withdrawal Bill, Sep 2017–Jun 2018, 4,437 speeches. Variables: date, speaker, party, word count, text.
- **Eurydice:** EU descriptions of 43 European education systems, collected around 2022, 4,465 chapters. Variables: country, chapter, text, url.
- **US Presidential Debate 2020:** first Trump–Biden debate, 29 Sep 2020, 789 speaking turns. Variables: speaker, minute, text.
- **ParlSpeech**: Full text corpora of parliamentary speeches in Austria, the Czech Republic, Germany, Denmark, the Netherlands, New Zealand, Spain, Sweden, and the United Kingdom

- … more. Ask me if you need help!

Pitches

Thank you very much!

Questions: milan.thies@unil.ch

Unil.